Richard Allen Doss

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Objective

Information Technology undergraduate seeking an AI / ML or software engineering internship. I build full applications end to end: backend APIs, LLM and agent workflows, data pipelines and deployed front ends. Six projects shipped or prototyped, most with public repos and deployed apps.

Education

Sri Sairam Engineering College, Chennai B.Tech, Information Technology (August 2025 – June 2029) | CGPA: 9.35/10

Alwin International Public School, Class XII, CBSE: 458/500 (91.6%)

Alwin International Public School, Class X, CBSE: 454/500 (90.8%)

Technical Skills

Languages: Python, Java, C, SQL

AI & Generative AI: LLM integration, Gemini API, tool-calling agents, RAG, agentic workflows, OpenRouter, local LLMs (Ollama)

Backend: FastAPI, Spring Boot, REST APIs

Frontend: React, Vite

Data & Databases: PostgreSQL, MySQL, Supabase, Google Earth Engine

Cloud & Tools: AWS EC2, Vercel, Git, GitHub, Postman

AI-Assisted Development: designing architecture, writing task-scoped prompts for coding agents, and reviewing and testing the generated code

Projects

FinPilot — AI personal finance decision-support agent | GitHub

- Built the data and agent core: a Gemini tool-calling orchestrator whose tools query real database functions, so the model never computes numbers itself

- FastAPI backend with React/Vite UI; 65 tests passing on ingestion and 16/16 on the analysis layer, each verified end to end with real data

Canopy — Forest biomass and carbon-stock mapping | GitHub | Live app

- Pairs GEDI L4A biomass with Sentinel-1/2, NASADEM and ESA WorldCover data to build an ML training dataset for above-ground biomass estimation
- Fully deployed app (FastAPI backend + React/Leaflet front end) showing biomass, carbon stock and CO2 equivalent on a 250 m grid for the Nilgiri Biosphere Reserve, live on Vercel with the backend running

AI Teacher Studio — Adaptive AI teacher | GitHub

- Generates lesson plans, multilingual speech and AI teacher videos for interactive learning
- Video pipeline renders Manim scenes (text, circuit, chart and other templates) over teacher narration, driven by an AI planner that outputs topic JSON

Corridor — Blockchain-based cross-border remittance platform | GitHub | Live demo

- Team of three; I owned the Supabase account layer, the UI and the core backend
- Backend: Node/Express API with Supabase JWT, a transfer state machine, mock KYC, an AML rules engine, FX quotes and an audit log

Invoice AI — Invoice tracker with AI-drafted emails for a share market trader | GitHub

- Login, dashboard of past tracks and a multi-step form; the AI drafts the email and the user reviews it before sending
- Supabase database and auth; small OpenRouter model for drafting with a template fallback, tested on a 10-case evaluation harness

WhatsApp Poll Bot — Group poll automation | GitHub

- Built with whatsapp-web.js for a 65-member class group: !pending lists students who have not voted, !votes lists voters per option, and teachers are managed by tagging them in chat; hosted on AWS EC2

Leadership & Hackathons

- **Team Lead**, AI Agent Hackathon by ProductSpace (national level)
- **Team Lead**, Bharat Academix hackathon (national level)
- **Team Lead**, Orion 1.0 by Sathyabama Institute (national level)
- **Team Lead**, Hack for Social Cause (MY Bharat portal), with the Canopy biomass idea

Certifications

NPTEL

The Joy of Computing Using Python	IIT Madras	90%	Gold
Programming in Java	IIT Kharagpur	92%	Gold
Developing Soft Skills and Personality	IIT Kanpur	93%	Gold
UN Sustainable Development Goals	IIT Hyderabad	94%	Gold (Top 1%)
Computer Architecture and Organization	IIT Kharagpur	71%	Elite
Ethics in Engineering Practice	IIT Kharagpur	73%	Elite

Coursera

Introduction to SQL (University of Michigan); Introduction to Hardware and Operating Systems (IBM); Introduction to Probability and Data with R (Duke University); Introduction to Software Product Management (University of Alberta); Strategic HRM: Talent, Performance, Compensation Design (Starweaver)

Coding Practice

Regular practice on SkillRack in Python, Java and C.

- **Java:** deep command of core Java and OOP, collections, multithreading and Spring Boot; NPTEL Programming in Java, 92% (gold)
- **Python:** strong in the language, using FastAPI to build backend APIs and AI projects; NPTEL Joy of Computing Using Python, 90% (gold)
- **C:** comfortable across the language, including pointers, dynamic memory, structures, file handling and linked lists
- **Data structures & algorithms:** arrays, strings, linked lists, stacks, queues, trees, graphs, hashing, sorting, recursion, dynamic programming and BFS, in Python and C

Declaration

I hereby declare that all the information given above is true and correct to the best of my knowledge.

Place: Chennai

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